SmolVLM: Redefining Small and Efficient Multimodal Models

Compact vision-language models for competitive performance under stri ts

Authors: Hugging Face & Stanford

Audience: ML researchers, on-device AI engineers, graduate students

SmolVLM

Architecture

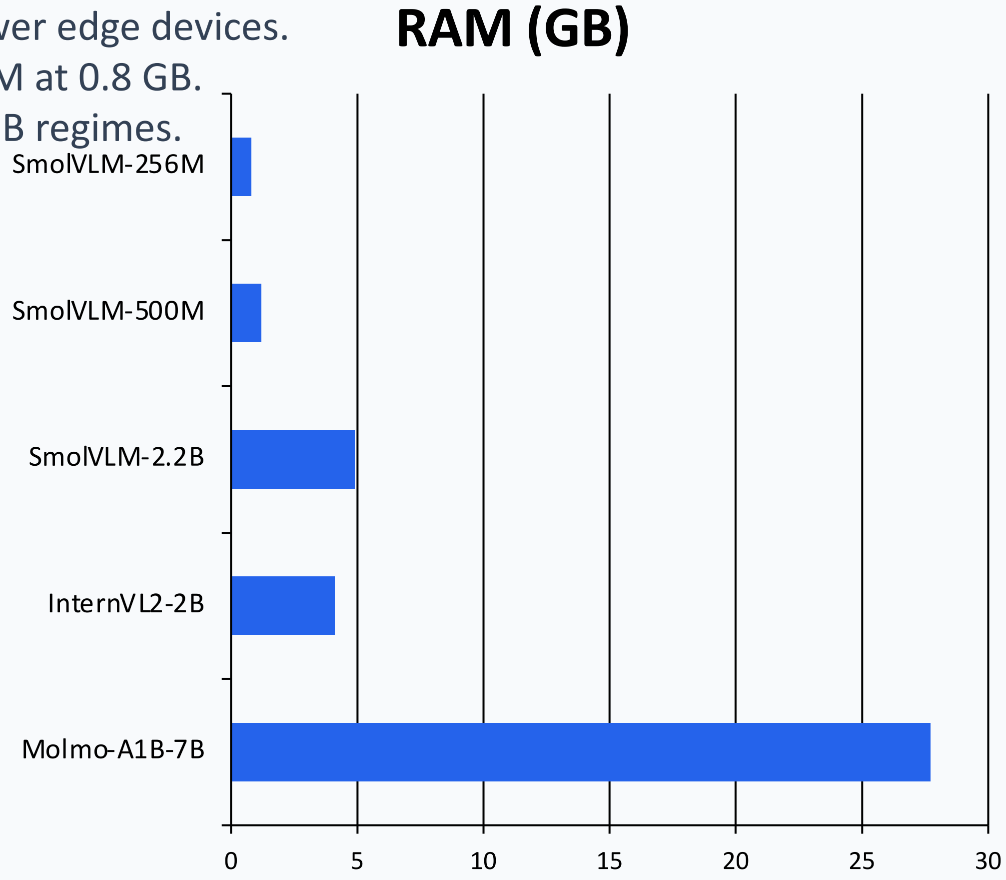
Large VLMs Are Powerful but Impractical for Edge Deployment

State-of-the-art VLMs deliver strong accuracy but demand large memory and compute.

High RAM requirements block deployment on smartphones and low-power edge devices.

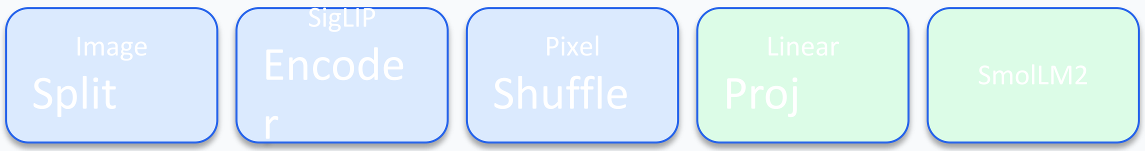
Example gap: Molmo-A1B-7B needs 27.7 GB GPU RAM vs SmolVLM-256M at 0.8 GB.

Objective: retain multimodal quality while pushing inference into sub-1GB regimes.



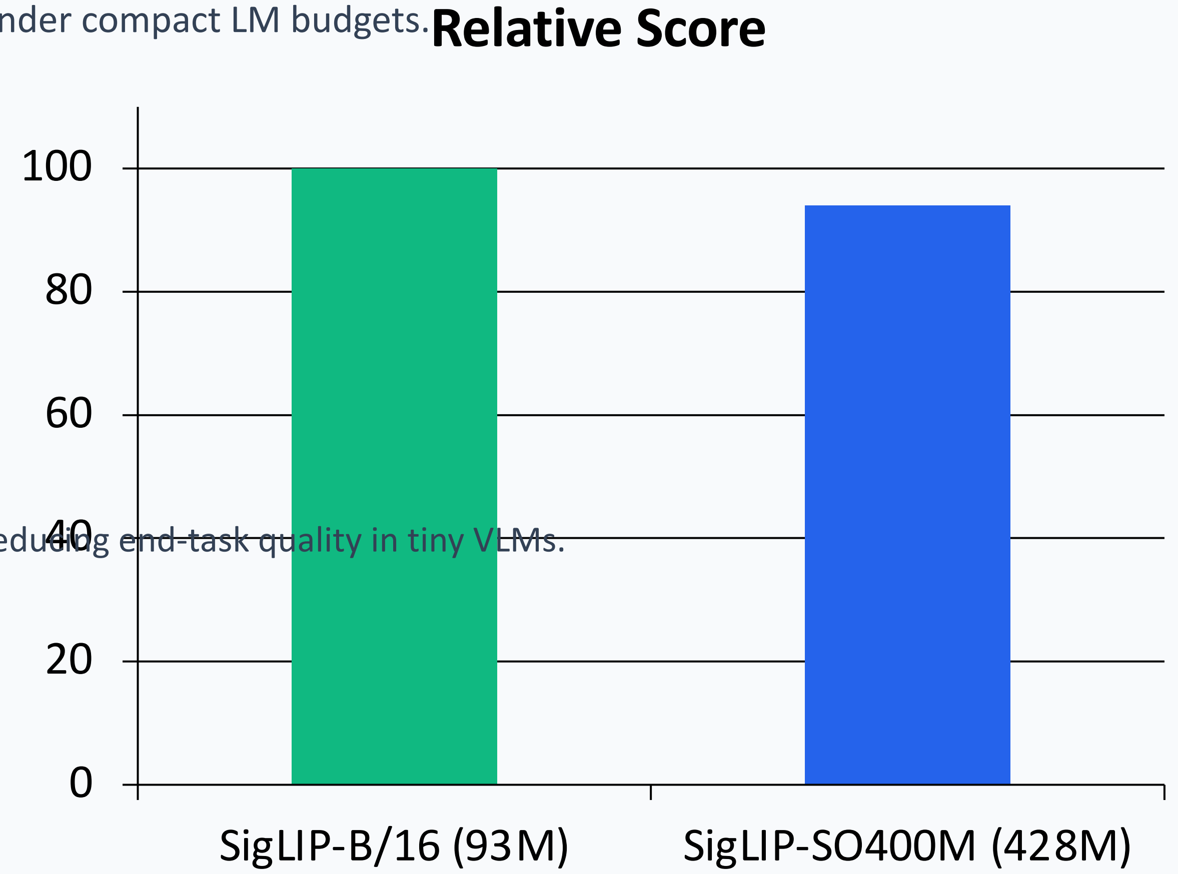
SmolVLM Architecture: Image Splitting, Pixel Shuffle, and SmolLM2 B

1) Input image is split into local crops + global view.
2) 93M SigLIP-B/16 vision encoder extracts visual tokens.
3) Pixel shuffle compresses tokens before projection.
4) Linear projection aligns vision features to SmolLM2 token space.
5) SmolLM2 decodes multimodal context for text generation.



Smaller Vision Encoders Outperform Larger Ones in Compact VLMs

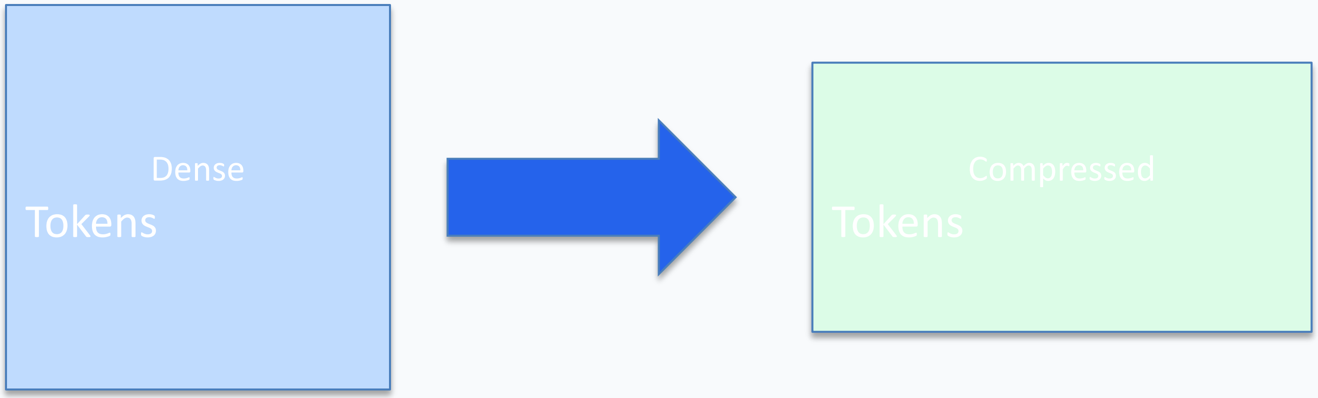
Finding 1: For the smallest language models, balanced compute allocation beats oversized vision towers.
93M SigLIP-B/16 encoder outperforms 428M SigLIP-SO400M under compact LM budgets.



Interpretation: Oversizing the encoder can starve the LM budget, reducing end-task quality in tiny VLMs.

Aggressive Pixel Shuffle (r=4) Cuts Visual Tokens 16× for Small Model

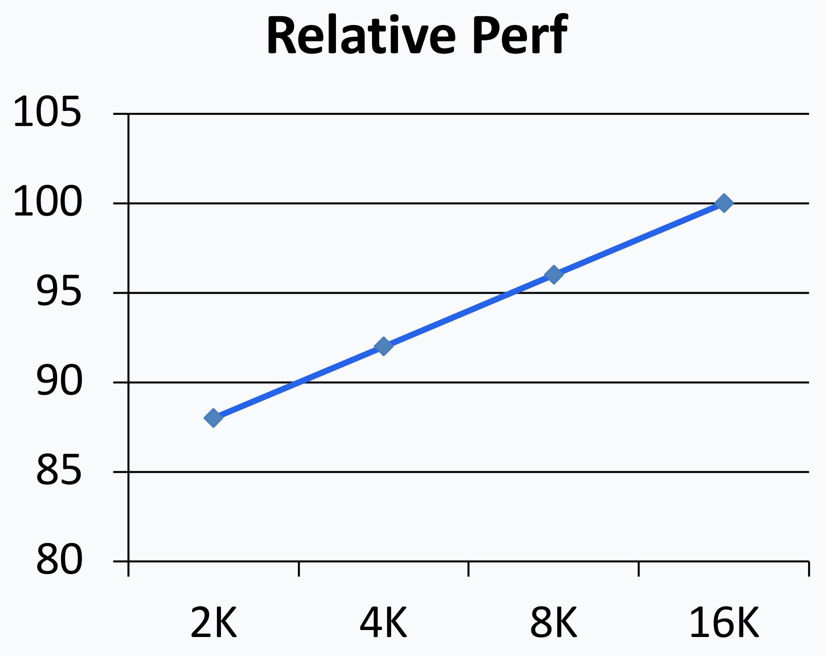
Pixel shuffle reorganizes spatial features to reduce token sequence length before fusion with language tokens.



r=4 ⇒ 16× token reduction
Best for smallest models
Larger models often prefer r=2

Extending Context from 2K to 16K Tokens Unlocks Higher Image Reso

Finding 2: Increasing context length significantly improves multimodal performance.
RoPE base adjusted from 10K to 273K to stabilize long-context extrapolation.
Fine-tuned on mixed short/long context data to preserve short-context quality.
SmolVLM-2.2B supports 16K context; 256M/500M variants use 8K context.
Longer context enables higher-resolution visual understanding and richer prompts.



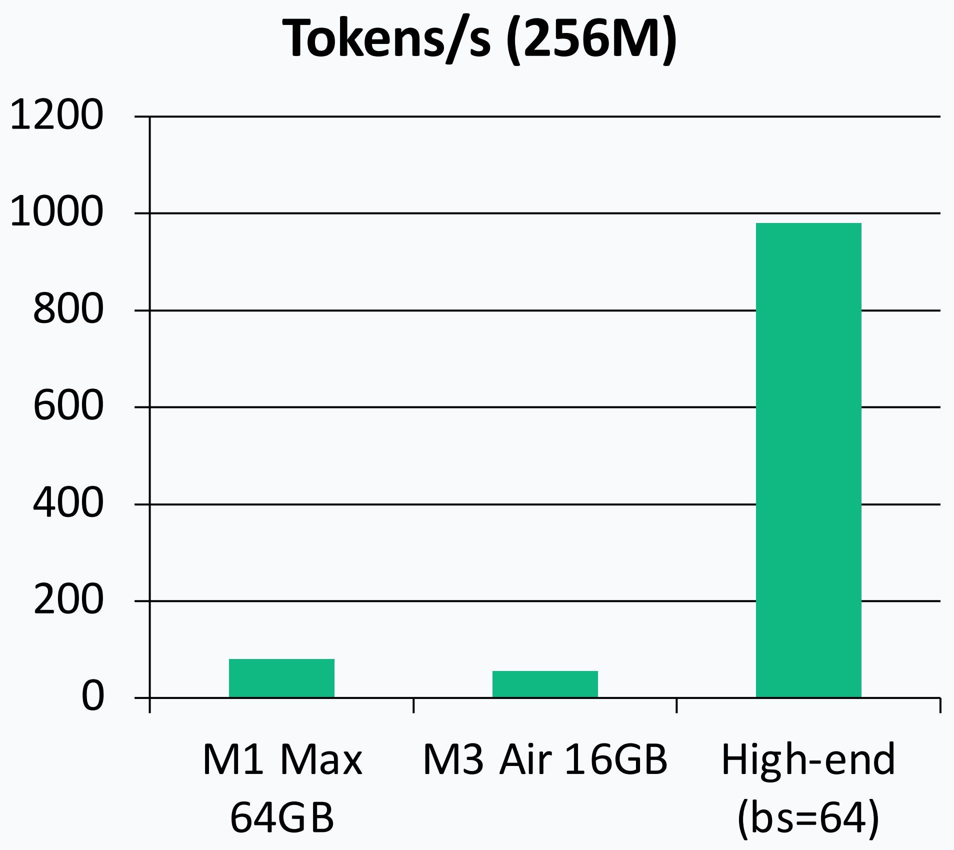
SmolVLM Achieves Competitive Performance at Sub-1GB to 4.9GB RA

Model	Avg Score	GPU RAM (GB)	Note
SmolVLM-256M	44.0%	0.8	Sub-1GB; surpasses Idefics-80B
SmolVLM-500M	51.0%	1.2	Strong mid-size tradeoff
SmolVLM-2.2B	59.8%	4.9	Rivals larger efficient OS models
InternVL2-2B (Best Efficient OS)	56.0%	4.1	Higher memory than 500M
Molmo-A1B-7B (Best Efficient OS)	N/A	27.7	Far above edge memory budgets

SmolVLM family scales quality with modest memory growth (0.8→4.9 GB), while maintaining competitive benchmark performance across single-image, multit

Real-Time Inference on iPhones: Up to 80 Tokens/Second on Apple Si

HuggingSnap app runs fully on-device (no cloud dependency).
SmolVLM-256M reaches ~80 tok/s on M1 Max 64GB.
At batch size 64, 256M reaches nearly 1000 tok/s on high-end Apple Silicon.
On MacBook Air M3 16GB, 256M still delivers ~55 tok/s.



Key Takeaways: Design Principles for Efficient Multimodal Models

Balance compute allocation across vision encoder and LM; bigger encoders are not always better.

Use aggressive token compression (pixel shuffle r=4) for the smallest models.

Scale context length (2K→8K/16K) with RoPE retuning for higher-resolution understanding.

Sub-1GB multimodal inference is practical with careful architecture co-design.

Open-source models, data, code, and app accelerate reproducible on-device AI research.

Implication: Efficient multimodal models can shift VLM capabilities from cloud-only to truly personal edge devices.